

Due: Thursday April 28th, latest at 2PM

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1. An electromagnetic field is described by the scalar potential V and vector potential $\mathbf{A}$, given by

$$V = \frac{2B_0}{\mu_0\epsilon_0}t; \quad \mathbf{A} = B_0 \left[x\hat{i} + (y \sin \omega t - 3z)\hat{k} \right]$$

where B_0 and ω are positive constants.

- (a) [6 pts] Find the electric and magnetic fields in this system.
 - (b) [6 pts] Show that Maxwell's third equation is satisfied.
 - (c) [4 pts] Find the displacement current density $\mathbf{J}_D$.
 - (d) [4 pts] Find the current density $\mathbf{J}$.
2. (a) [5 pts] Maxwell's first equation in differential form is an expression for the divergence of the electric field. Rewrite this equation in terms of the scalar and vector potentials, V and $\mathbf{A}$.
- (b) [3 pts] Write down the Coulomb gauge condition for the divergence of the vector potential. Use this condition to express Maxwell's first equation as an equation for the scalar potential V .
- (c) [4 pts] Write down the Lorenz gauge condition. Use this condition to express Maxwell's first equation as an equation for V .
3. Two metallic plates are placed parallel and close to each other. They each have area A . They carry opposite charges $+Q$ and $-Q$. Such an arrangement is known as a *capacitor*.
- (a) [5 pts] The plates are large enough that, if we are considering the region between the two plates (far from the plate edges), we can regard each charged plate as an infinite sheet of charge. Show that the magnitude of the electric field in the region between the plates is

$$E = \frac{Q}{A\epsilon_0}$$

- (b) **[6 pts]** The *capacitance* of a capacitor is the charge on a plate divided by the potential difference between the two plates ($C = Q/V$). If the distance between the plates is d , calculate the capacitance of the arrangement.
- (c) **[7 pts]** Next, consider the situation in which capacitor is *discharging*, i.e., the magnitude of the charge on the each plate is decreasing as $Q(t) = Q_0 e^{-t/T_0}$. Find the displacement current density in the region between the two plates.